YI-JYUN SUN

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EDUCATION

University of Illinois Urbana-Champaign (UIUC) *Illinois, U.S.*

M.S. in Computer Science (Thesis), Advisors: Dilek Hakkani-Tur & Gokhan Tur GPA: 3.96 / 4.0 Aug 2025 –Present

National Yang-Ming Chiao-Tung University (NYCU) *Hsinchu, Taiwan*

B.S. in Computer Science, GPA: 3.61/4.0 (Last 60 GPA: 3.94/4.0) Sep 2020 – June 2025

PUBLICATIONS

[1] **Y.-J. Sun**, D Hakkani-Tur, S Syed, J Rehg, "Structuring Matters: Insurance Policy Formalization for Agentic Decision-Making." Under review at NAACL 2027.

[2] NK Hoang, S Mehri, T.-A. Hsu, **Y.-J. Sun**, QXN Truong, KD Doan, D Hakkani-Tür, "PSI-Bench: Interpretable and Clinically Meaningful Evaluation of Depression Patient Simulators." SIGDIAL Social Sim Workshop, 2026. Best Paper Award.

[3] S. Jhunjhunwala, **Y.-J. Sun**, A Deora, Z Wang, NB Shankar, T Christ, C Wang, A Alwan, D Hakkani-Tür, "The effect of transcription quality on LLM-judge scoring: a case study of ASR mediated children QA grading." Under review at ICASSP 2027.

[4] Y Li, **Y.-J. Sun**, V Nguyen, K Unnikrishnan, A Agrawal, D Hakkani-Tür, "From Tacit Understanding to Downstream Collaboration." In preparation.

[5] S Dey, **Y.-J. Sun**, D Hakkani-Tür, G Tür, "Know Your Mistakes: Towards Preventing Overreliance on Task-Oriented Conversational AI Through Accountability Modeling." ACL, 2025.

[6] **Y.-J. Sun**, S Dey, D Hakkani-Tür, G Tür, "Confidence Estimation for LLM-based Dialogue State Tracking." IEEE Spoken Language Technology Workshop (SLT), 2024.

WORK EXPERIENCES

ConvAI Lab - University of Illinois Urbana-Champaign *Illinois, U.S.*

Graduate Research Assistant Aug 2025 – Present

- Developing an agentic AI system to automate clinical prior authorization workflows through insurance policy understanding, patient-specific compliance assessment, and interactive clinician support.
- CELaRAI [Website]: Developing LLM-based comprehension question generation and automated assessment systems for an interdisciplinary initiative advancing personalized early literacy education.

Amazon - One Medical Product Dev Team *California, U.S.*

Applied Scientist Intern May 2026 – Aug 2026

- Fine-tuned and evaluated multiple smaller language models using SFT and GRPO for the healthcare triage task, achieving ~7% higher accuracy than a closed-weight foundation model while reducing inference latency by 3–5×.
- Improved model generalization to unseen clinical concepts by 5% and matched the foundation model's adaptability under modified task guidelines, validated across 1,500+ held-out augmented evaluation samples.

Omniguider Inc. (Foxconn Technology Group) *Taipei, Taiwan*

Machine Learning Intern Jul – Aug 2025

- Built a library recommender on 40K+ books using LLM profiling, embedding model, autoencoder, and clustering in PyTorch.
- Delivered personalized book recommendations through an interactive Streamlit web app supporting dynamic user interactions.

Big Data Mobile *Taipei, Taiwan*

AI Intern Feb – Jun 2025

- Developed an arrhythmia classification model in Tensorflow achieving 99% accuracy on MIT-BIH Arrhythmia ECG dataset.
- Built a Python ECG toolkit with NeuroKit2 to standardize segmentation, annotation, and resampling for automated pipelines.

Industrial Technical Research Institute *Hsinchu, Taiwan*

LLM Application Development Intern May – Dec 2024

- Built an AI tool for a news outlet to automate content extraction and article filling, boosting collection efficiency by 40%.
- Designed adversarial prompting techniques using rewriting and nesting modules to mask harmful behavior.

University of Illinois Urbana-Champaign *Illinois, U.S.*

Undergraduate Research Assistant May – Aug 2024

- Researched uncertainty estimation for dialogue state tracking, improving zero-shot JGA 8.5% and fine-tuned model AUC 20%.
- Proposed an accountability model for dialogue agents, improving dialogue state tracking JGA by 3% with accountability head.